

WHOSE STANDARD OF CARE? FOREIGN-GUIDELINE AND OUTDATED-VERSION DEVIATIONS OF LLMS FROM VIETNAMESE MINISTRY OF HEALTH GUIDELINES

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BACKGROUND: Students and health workers use large language models (LLMs) for clinical reference, but LLMs may answer with the values of foreign or superseded guidelines, whereas Vietnamese Ministry of Health (MoH) guidelines may differ in doses, thresholds and drugs.

OBJECTIVE: To build a method that measures how LLM answers deviate from current MoH guidelines and classifies each error by source (a named foreign guideline, a superseded MoH version, or no recorded source), and to pilot it on a locally run open model.

METHODS: Pilot on 65 recommendations with specific values quoted verbatim from official PDFs; 61 recommendations from 12 documents still in force were analysed. Quotations were checked by two separate, blinded AI passes (Claude) with an AI referee. Recommendations carry versioned WHO, US and European values (when available), superseded values (when available) and a rule-generated decoy value (when the rule allows) to control for chance matches. Qwen3-8B (quantized, run on a laptop) answered in Vietnamese and English under three conditions: no country cue, "according to the MoH", and with the correct guideline passage. Answers were graded by pre-specified value-comparison rules, and all labels were re-checked by two blinded AI passes with an AI referee.

RESULTS: For short answers, the rule-based grader matched the AI-adjudicated labels in 89.5% (kappa 0.83). Under the MoH cue in Vietnamese, on 20 recommendations whose MoH value differs from at least one named foreign guideline, the model was correct in 7 (35.0%; 95% CI 15.4%–59.2%), matched a foreign value in 2, matched no value in the reference set in 7, and 4 were graded as non-answers. On 15 recommendations with a decoy, foreign versus decoy matches were 2 versus 0 (short answers) and 5 versus 4 of 26 multiple-choice answers. On 20 recommendations with a superseded version, 3 answers matched the superseded value. On recommendations concordant with international guidelines, 9 of 21 were correct. With the

guideline passage, correct answers were 13 of 20 for conflict recommendations (9 improved, 3 worsened; $p = 0.15$) and 19 of 21 for concordant ones.

CONCLUSION: For this small open model, most errors were values matching no recorded source, with no sign of foreign defaults beyond the decoy (chance) level. Supplying the MoH text increased correct answers, but errors remained. These are preliminary results on a hand-picked sample, checked by AI without clinician review; the main study will test them across the guideline corpus and several locally run open models.

KEYWORDS: large language models; clinical practice guidelines; ministry of health; model evaluation; medical education

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